Simon Akhter

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Education

University of Waterloo Waterloo, Canada
Bachelor of Applied Science in Management Engineering *Expected Apr 2029*
Relevant Coursework: *Data Structures and Algorithms, Databases and Software Design, Distributed Systems*

Experience

Software Engineer Intern Sep 2026 – Present
University Health Network - Princess Margaret Cancer Centre *Toronto, ON*

- Building and redesigning UHN Research's intranet platform serving 6,000+ researchers, clinicians, trainees, and staff by developing features across React, Next.js, Python, MySQL, and dotCMS
- Optimizing MySQL data access by profiling query execution plans, eliminating N+1 queries, designing composite indexes, and refactoring joins, reducing database query latency by 60%
- Designing backend data models and transactional workflows with ACID guarantees, foreign key constraints, and automated migrations; developed 15+ REST APIs reducing API response latency by 40%

Software Engineer Intern Feb 2026 – May 2026
Zomp *Toronto, ON*

- Built a self-hosted documentation CMS supporting 10+ documentation sites with automated GitHub-based publishing, Docker deployments, and Azure Container Apps, reducing content deployments by 96%
- Implemented an automation layer handling role-based access control, file uploads, spell-check validation, and atomic GitHub commits across 10+ documentation sites, reducing manual deployment steps from 8 to 1
- Designed CI/CD pipelines using GitHub Actions, Docker, Azure Container Registry, and Azure Container Apps to automate image builds, registry pushes, and production validation with automated deployment validation

Software Developer Intern May 2025 – Aug 2025
Code Ninjas *Richmond Hill, ON*

- Designed and deployed an interactive check-in portal across 2 Code Ninjas locations, enabling live student tracking and reporting for 200+ families through a React, Node.js, Express.js, and MongoDB application
- Enhanced API throughput by 25% by migrating CPU-intensive tasks to Node.js worker threads, preventing event loop blocking and improving backend responsiveness during concurrent check-in and data processing workflows
- Shipped 8 production features including real-time notifications, uploads, and reporting tools, reducing feature delivery cycles from 2 weeks to 5 days

Projects

Market Intelligence | *Go, B+ Trees, Transactions, REST APIs, Docker, GitHub Actions* Jun 2026

- Built a database engine in Go with copy-on-write B+ trees, atomic transactions, indexing, and crash recovery; optimized page splitting to reduce insertion latency 77% and allocations 85% versus the baseline implementation
- Built an indexed time-series pipeline with validated OHLCV ingestion, point-in-time features, and reproducible market analysis across five years of AAPL data
- Built REST APIs, durable background jobs, authentication, and observability tooling; benchmarked the system at 262 requests/sec with 93 ms p95 latency and automated race, fuzz, and crash-recovery testing through Docker CI

DocuEdit | *React, Node.js, WebSockets, Yjs, MySQL* Jun - Jul 2026

- Built a real-time collaborative documentation editor with React, Node.js, MySQL, and WebSockets, supporting concurrent document editing with persistent document storage
- Implemented concurrent editing with Yjs CRDT synchronization over WebSockets, enabling edits to converge across connected clients and preserving state across reconnects
- Built a VS Code extension for importing files, folders, and workspaces into editable documentation while implementing session authentication and document-level editor and viewer authorization

Skills

Languages: Go, Python, TypeScript, JavaScript, C++, Java, C#, SQL
Frameworks: React, Next.js, Node.js, Express, FastAPI, Flask, ASP.NET
Developer Tools: MySQL, PostgreSQL, MongoDB, SQLite, Docker, Git, GitHub Actions, AWS, Azure, Linux